

In today's class, we'll build on our foundation of Linear Regression and dive deeper into the **mathematics, implementation, and evaluation** of regression models.

Key Topics You'll Learn

1. Errors & Residuals

- Understand what residuals are and why we measure them.
Learn why simply summing errors doesn't work (positive and negative values cancel out).
Introduction to **Mean Squared Error (MSE)** as a more reliable error metric.

2. Ordinary Least Squares (OLS)

- Explore why we square errors in regression.
Understand how OLS minimizes the overall error and helps in fitting the "best line" to data.

3. Gradient Descent Optimization

- Learn how algorithms *iteratively* find the best parameters (slope & intercept) instead of guessing.
Build intuition about the optimization process behind modern ML models.

4. Univariate Linear Regression with Scikit-learn

- Move from manual intuition to practical implementation using Python's Scikit-learn library.
- Understand how to fit, train, and interpret a simple regression model.

5. Model Evaluation – R^2 (Coefficient of Determination)

- Go beyond error metrics and learn how well your model explains the variation in the data.
- Interpret R^2 in practical terms: when is it high, when is it misleading?

6. Multivariate Linear Regression

- Extend linear regression from a single predictor (like car's max power → price) to multiple predictors (e.g., mileage, age, horsepower together).

- Understand how coefficients and intercepts represent relationships in higher dimensions.
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Why This Matters

- **Critical Thinking:** You'll see *why* we choose MSE over other error functions.
- **Hands-On Skills:** Learn how to implement regression models using Scikit-learn, a must-know library for data science roles.
- **Evaluation Mindset:** By learning R^2 , you'll start thinking about model quality, not just fitting.
- **Scalability:** Understanding multivariate regression prepares you for real-world data where outcomes depend on many factors simultaneously.

By the end of this session, you'll not only be able to **fit and evaluate linear regression models** but also build intuition for how optimization drives all of machine learning